

Reviewer Pack — Minimal Rank of Tropicalized Quantum Entanglement Bounds AC^0 ...

Entry 8f0db998e52a · INCONCLUSIVE

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Minimal Rank of Tropicalized Quantum Entanglement Bounds AC^0 Circuit Depth

Entry ID: 8f0db998e52a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 03:24:57 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory (Tropicalization) **Field B** (complexity object): Complexity Theory: AC^0 Circuit Depth

Statement:

{‘text’: ‘For every AC^0 circuit C with n inputs, the minimal rank of its tropicalized quantum entanglement, $\psi(C)$, is upper bounded by a function $f(n) = O(\log^2 n)$. Furthermore, for any instance of PARITY computation on n variables, there exists an AC^0 circuit C computing it such that $\psi(C) = \Theta(\log n)$.’, ‘counterexample’: ‘An AC^0 circuit C for PARITY computation with $\psi(C) < \Theta(\log n)$.’}

Rationale (proposer’s reasoning):

{‘text’: ‘Tropicalization is a powerful tool in quantum information theory that provides a bridge between abstract algebraic structures and physical processes. By studying the minimal rank of tropicalized quantum entanglement, we may uncover non-trivial relationships between quantum algorithms and classical complexity classes like AC^0 .’}

Taxonomy category: AC^0_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ecff3879c03be37a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all PARITY instances with n variables and 30 random seeds, the minimal rank of tropicalized quantum entanglement $\psi(C)$ for the constructed AC^0 circuit C satisfies $\psi(C) \leq O(\log^2 n)$, and for at least one seed, $\psi(C) = \Theta(\log n)$. The conjecture is falsified if any seed produces a $\psi(C) < \Theta(\log n)$ or $\psi(C) > O(\log^2 n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - tropicalization AND quantum entanglement AND AC0 circuit depth - quantum information theory AND minimal rank tropicalized AND AC0 complexity - AC0 circuit depth AND PARITY computation AND tropicalized quantum entanglement bound

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i
            for j in range(i+1, m):
                if abs(A[j][i]) > abs(A[max_row][i]):
                    max_row = j
            A[i], A[max_row] = A[max_row], A[i]
            pivot = A[i][i]
            for j in range(n):
                A[i][j] /= pivot
            for j in range(m):
                if j != i:
                    factor = A[j][i]
                    for k in range(n):
                        A[j][k] -= factor * A[i][k]
        return A

    def matrix_multiply(A, B):
        m, n = len(A), len(B[0])
        p = len(B)
        C = [[0] * n for _ in range(m)]
        for i in range(m):
            for j in range(n):
```

```

        for k in range(p):
            C[i][j] += A[i][k] * B[k][j]
    return C

def tropical_rank(matrix):
    rank = 0
    matrix_tropical = [[max(a, b) for a, b in zip(row, col)] for row, col in zip(*matrix)]
    reduced_matrix = gaussian_elimination(matrix_tropical)
    for row in reduced_matrix:
        if any(x != float('-inf') for x in row):
            rank += 1
    return rank

n = random.randint(5, 40)
A = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]

rank = tropical_rank(A)
instances_tested = 1
conjecture_holds = rank <= math.log2(n) ** 2 and (rank == math.log2(n) or rank == math.log2(n))
counterexample = "" if conjecture_holds else f"Rank {rank} does not satisfy bounds for n={n}"

return {
    "metric_name": "tropical_rank",
    "metric_value": rank,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else list(range(2, 100, 4))

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    std_rank = math.sqrt(sum((r["metric_value"] - mean_rank) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std={std_rank} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\\Rank {results[0]['metric_value']} does not satisfy bounds for n={n}")
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ab77564.py", line 80, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ab77564.py", line 61, in run_trial
    rank = tropical_rank(A)
            ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4ab77564.py", line 51, in tropical_rank
    matrix_tropical = [[max(a, b) for a, b in zip(row, col)] for row, col in zip(*matrix)]
                                ^^^^^^^^^
```

ValueError: too many values to unpack (expected 2)

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means we cannot verify if the conjecture is supported or falsified according to the pre-registered criteria. | next: Re-run the test without errors to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14739
2	preregistration	ollama_remote	glm4:latest	0	0	6261
3	novelty	ollama_remote	glm4:latest	0	0	4650
4	novelty	ollama_remote	glm4:latest	0	0	5595
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16224
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12036
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	28880
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13224
9	judge	ollama_remote	glm4:latest	0	0	49851

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 151460 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in [research/audit/8f0db998e52a.jsonl](#))

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8f0db998e52a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/8f0db998e52a.tar.gz` (if generated)